Expressive Performance Generation with VirtuosoNet

Transforming Schumann's Variation V from Etude to Adagio

Group: J.S. BACH

JOHANNES KEPLER UNIVERSITY LINZ

The Task: Expressive Rendering

Goal: Generate a human-like performance from a raw MusicXML score.

The Challenge: Computers play perfectly in time (robotic). We need:

- **Micro-timing (Rubato):** Speeding up/slowing down for emotion.
- **Dynamics (Velocity):** Loudness shaping.
- **Articulation:** Staccato vs. Legato.

The Constraint (Interpretation):

- **Original Score:** *Etude* (Fast, technical).
- **Project Requirement:** *Adagio* (Slow, lyrical, 60–90s).
- **Objective:** Force the model to fight its training bias.

The Model: VirtuosoNet (HAN)

Architecture: Hierarchical Attention Network (HAN).

Why HAN? Musicians think in structures, not linear notes.

Three Levels of Processing:

1. **Note Level (Micro):** Pitch, duration, local neighbors.
2. **Voice Level (Texture):** Separates Melody (freedom) vs. Accompaniment (stability).
3. **Measure Level (Macro):** Understands rhythmic grid and phrasing.

Training Data: Yamaha e-Piano Competition

Source: Yamaha e-Piano Competition Dataset.

Content: >1,000 performances by professional pianists.

Hardware: Recorded on **Yamaha Disklavier** (Acoustic-Electric Hybrid).

Key Advantage: Captures **High-Fidelity Physics**.

- Exact hammer velocity.
- Precise key release timing.

Learned Embeddings: The model learned distinct "Composer Styles" (e.g., Schumann vs. Chopin).

The Problem: Baseline Failure

Experiment 1: Zero-Shot Inference (Default Settings).
Embedding: Schumann (Matches the composer of the piece).
Analysis: Model saw dense 16th notes → Predicted "Virtuoso Etude".
Result:

- **Tempo:** ~160 BPM (*Vivace*).
- **Duration: 39 Seconds**.
- **Quality:** Mechanical, "Machine Gun" effect.

Outcome: Failed the *Adagio* constraint.

The Solution: The "Chopin" Strategy

Technique: Latent Space Manipulation (Style Transfer).
Hypothesis:

- **Schumann Embedding:** Biased toward rhythmic rigidity.
- **Chopin Embedding:** Associated with **Rubato** and **Cantabile** (Singing style).

The Fix:

1. **Switch Embedding:** Schumann → Chopin.
2. **Force Constraint:** Hard Tempo limit = **60 BPM**.

Effect: The model fills the slow tempo with expressive timing (rubato) rather than robotic silence.

Final Results

Quantitative Success:

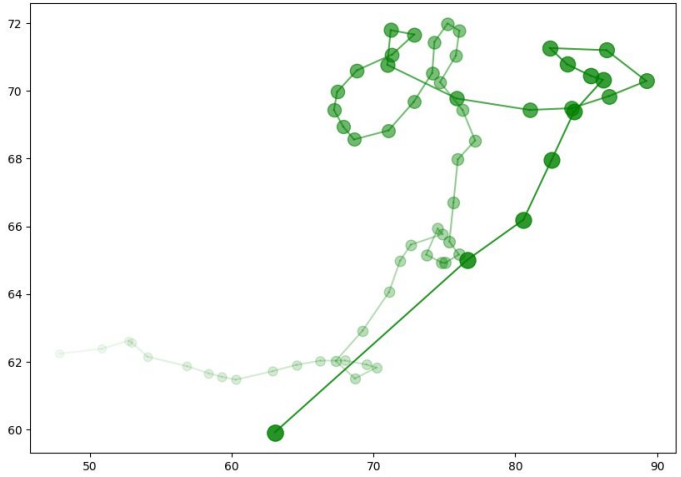
- **Duration:** 39s → **63s** (Met the 60-90s goal).
- **Tempo:** Stable at 60-72 BPM.

Qualitative Success:

- Human-like "breathing" in phrases.
- Dynamic contrast (Velocity 48–90).

Visual Evidence:

- *The "Worm" plot showing Tempo vs. Dynamics trajectory.*



Music Generation: Creative Process

Goal: compose an original piano piece for live Disklavier performance
Followed a human-in-the-loop composition process:

- Used Hookpad to explore chord progressions and structure
- AI suggestions guided harmony, human decisions shaped melody and form

Focuses on:

- Loopable structure
- Clear harmonic motion
- Piano-centric texture

Music Generation: Inspiration

- Inspired by video game and fantasy background music
- Focus on music that supports atmosphere rather than dominating attention
  Influences include:
    - Repetitive but evolving harmonic patterns
    - Calm, exploratory melodic motion
- Designed to feel:
    - Loopable
    - Immersive
    - Emotionally consistent over time

Conclusion

Summary: SOTA models are not just "Black Boxes."

Key Insight:

- **Hierarchy (HAN)** allows for structural coherence at slow tempos.
- **Latent Space (Embeddings)** allows for creative steering.

Takeaway: We successfully used domain knowledge (Music History) to contradict training bias and achieve a specific artistic goal.